TSM 099

Homework 2

Polynomial structure and operations

Topics 9-12 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Work without a calculator before checking. Give exact, simplified answers.

Terms, factors, and exponents

1. Simplify. Keep unlike terms separate.

$7x^2 - 3x + 4 - 2x^2 + 8x - 9$

2. Simplify. Keep unlike terms separate.

$3ab - 5a^2 + 7ab + 2a^2$

3. Simplify. Keep unlike terms separate.

$(-2x^3y)(5xy^2)$

4. Simplify. Keep unlike terms separate.

$(3a^2b)^2$

Read the structure

5. Simplify (2x) cubed and 2x cubed. Evaluate each at x = -1 and explain the role of parentheses.

Distribute and combine

6. Expand as needed and combine like terms.

$-3x^2(2x^3 - x + 4)$

7. Expand as needed and combine like terms.

$2(3x - 4) - 5(x + 1)$

8. Expand as needed and combine like terms.

$(4x^3 - 2x^2 + x) - (x^3 + 5x^2 - 3)$

9. Expand as needed and combine like terms.

$3(x - 2) - 2(4 - x) + 5x$

Binomial products

10. Multiply and combine like terms.

$(3x - 2)(2x + 5)$

11. Multiply and combine like terms.

$(4x + 1)(x - 3)$

12. Multiply and combine like terms.

$(2x - 5)^2$

13. Multiply and combine like terms.

$(3x + 4)(3x - 4)$

14. Multiply and combine like terms.

$(2a - 3b)(a + 4b)$

Explain and verify

15. Simplify 2(x + 1) squared - (x - 2)(x + 2). Then check at x = -1.

16. A student writes (x - 5) squared = x squared + 25. Explain the missing term by showing distribution, and correct the result.